

CODE:

```
#include <LiquidCrystal.h>

#include "DHT.h" // Use Adafruit's DHT library

// LCD Pins

const int rs = 13, en = 12, d3 = 11, d4 = 10, d5 = 9, d6 = 8; LiquidCrystal lcd(rs, en, d3, d4, d5, d6);

// Pin Definitions int pump = 4;

int fan = 5; int light = 6; int ldr = 2;

int moist = A0;

#define DHTPIN 3 // Pin for DHT sensor

#define DHTTYPE DHT11 // Sensor type: DHT11 or DHT22 DHT dht(DHTPIN, DHTTYPE);

void setup() { Serial.begin(9600);

// Pin Modes pinMode(ldr, INPUT);

pinMode(pump, OUTPUT); pinMode(fan, OUTPUT);

pinMode(light, OUTPUT);

pinMode(moist, INPUT);

// Initialize LCD lcd.begin(16, 2);

lcd.clear(); lcd.setCursor(0, 0); lcd.print("Green House"); lcd.setCursor(0, 1); lcd.print(" Monitoring ");

Serial.println("Green House Monitoring"); delay(1000);

digitalWrite(light, LOW); digitalWrite(fan, LOW); digitalWrite(pump, LOW); dht.begin(); // Initialize DHT
sensor

}

void loop() { DHT_CHECK(); MOISTURE(); LDR_CHECK();

}

void LDR_CHECK() {

if (digitalRead(ldr) == HIGH) { lcd.clear();

lcd.setCursor(0, 0);

lcd.print("Low Light"); Serial.println("Low Light"); digitalWrite(light, HIGH); delay(1000);

} else { lcd.clear();

lcd.setCursor(0, 0); lcd.print("Normal Light");

Serial.println("Normal Light");

digitalWrite(light, LOW); delay(1000);
```

```
}  
  
void MOISTURE() {  
  
int moistval = analogRead(moist); lcd.clear();  
  
lcd.setCursor(0, 0);  
  
lcd.print("Moist:" + String(moistval)); delay(800);  
  
if (moistval > 600) {  
  
Serial.println("Moisture: " + String(moistval)); lcd.clear();  
  
lcd.setCursor(0, 0); lcd.print("Low Moisture"); Serial.println("Low Moisture"); digitalWrite(pump, HIGH);  
delay(1000);  
  
} else {  
  
digitalWrite(pump, LOW); delay(1000);  
  
}  
  
}  
  
void DHT_CHECK() {  
  
float humidity = dht.readHumidity();  
  
float temperature = dht.readTemperature(); if (isnan(humidity) || isnan(temperature)) {  
  
Serial.println("Failed to read from DHT sensor!"); return;  
  
}  
  
Serial.print("HUMIDITY: "); Serial.print(humidity); Serial.println("%"); Serial.print("TEMPERATURE: ");  
Serial.print(temperature); Serial.println("C");  
  
lcd.clear(); lcd.setCursor(0, 0); lcd.print("HUMIDITY:"); lcd.setCursor(10, 0);  
  
lcd.print(humidity); lcd.setCursor(0, 1); lcd.print("TEMP:"); lcd.setCursor(6, 1); lcd.print(temperature);  
delay(1000);  
  
if (temperature > 35) {  
  
lcd.clear(); lcd.setCursor(0, 0); lcd.print("High Temp"); digitalWrite(fan, HIGH); delay(1000);  
  
} else {  
  
digitalWrite(fan, LOW); delay(1000);  
  
}  
  
}
```